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Instrumentation, Telemetry, and Spear Recovery Systems

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1. On-Spear Instrumentation Suite

All sensors use Aegis Telluric-Proof architecture: PMMA optical fiber, FBG sensing, LoRa wireless.

1.1 Sensor Package

Sensor	Type	Range	Sample Rate	Mass (g)
3-axis accelerometer	MEMS, fiber-coupled	±500 g	10 kHz	2
Angular rate (gyro)	MEMS, fiber-coupled	±10,000 deg/s	1 kHz	3
Static pressure	FBG optical	0.001-110 kPa	100 Hz	4
Dynamic pressure (pitot)	FBG optical	0-5 MPa	100 Hz	3
Nose temperature	ZrB2-SiC embedded FBG	-50°C to 3000°C	10 Hz	2
GPS	Ceramic-packaged chip	all	10 Hz	5
LoRa transmitter	Ceramic chip	915 MHz, 250 mW	—	4
Camera	ZrO2-housed, fisheye	4K video	60 fps	8
MRAM data storage	Radiation-hard	4 GB	—	3
Power (LiFePO4)	ZrO2-housed cell	50 Wh	—	180

Total instrument mass: ~214 g (fits in 60mm diameter × 150mm payload section)

1.2 Telemetry Link Budget

LoRa 915 MHz link from 100 km altitude to surface vessel: Free-space path loss: L_fs = 20*log(4*pi*d/lambda) = *20*log(4*pi*100000/0.326) = 141 dB LoRa transmitter EIRP: 250 mW → +24 dBm LoRa sensitivity (spreading factor 12): -137 dBm Link margin: +24 - 141 - (-137) = **+20 dB** — excellent link at 100 km altitude.

Real-time data at ~500 bps (LoRa SF12): pressure, temperature, acceleration → trajectory reconstruction.

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2. Recovery Systems

2.1 Parachute Deployment (Suborbital Missions)

For suborbital flights reaching 100 km apogee, spear re-enters at: v_reentry = sqrt(2 * 9.81 * 100000) = 1,407 m/s (vacuum equivalent) With atmospheric braking at 80 km: v at 80km = ~1,200 m/s

KNbO3-BaTiO3 mortar parachute deployment at 5 km altitude: - Deployment triggered by barometric threshold (FBG pressure sensor reads 54 kPa) - KNbO3-BaTiO3 linear actuator ejects parachute canister from nose - Pilot chute deploys: 0.5 m diameter, extracts main drogue - Main chute: 3 m diameter ribbon parachute (suitable to 300 m/s deployment) - Final descent rate: v_term = sqrt(2mg / (rho*A*Cd)) = sqrt(2*2*9.81/(1.225*7.07*1.2)) = 2.0 m/s

Splashdown velocity: 2.0 m/s — safe for recovery.

2.2 GPS + LoRa Tracking

LoRa beacon remains active throughout flight and post-splashdown (waterproof ZrO_2 housing). Signal range on ocean surface (flat terrain, LoRa SF12): 15 km Vessels monitor beacon and dispatch recovery RIB to splashdown coordinates.

2.3 Recovery Vessel Specification

Fast recovery RIB: 8m length, BFRP hull, outboard engine (35 kW). Speed: 45 knots → recovers spear within 3 km in < 5 minutes, 50 km in < 75 minutes.

Citations (Vol 25)

- LoRa Alliance, LoRaWAN Physical Layer specification, 2022
- CSMFAB000000000013 V2.0, PMMA optical fiber instrumentation, CSM 2026
- Knacke, T.W., Parachute Recovery Systems Design Manual, NWC TP 6575, 1992
- AeroTech Parachute Systems, descent rate calculations, 2024

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